

Autonomous Rescue & Extraction Drone

Problem statement: Rescue teams struggle to reach victims safely and quickly during severe floods or landslides in urban areas due to debris and adverse weather conditions. Hence, rescue operations are delayed, and mortality rates are increased.

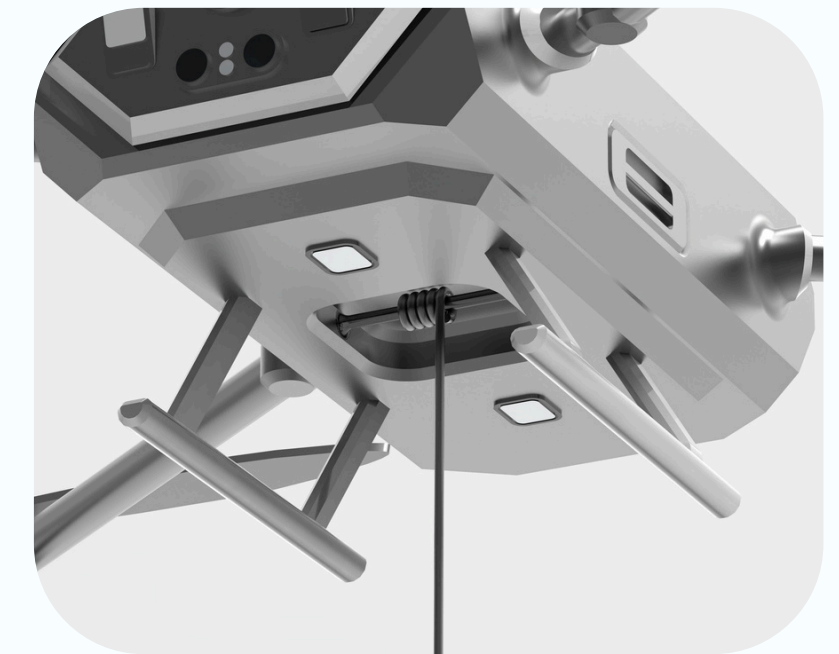
Our solution: Autonomous Rescue & Extraction Drone (A.R.E.D.) is an AI-powered octocopter drone capable of autonomously locating victims, deploying a smart human harness, and safely guiding them to safety, at the same time providing immediate medical diagnosis.

Key components:

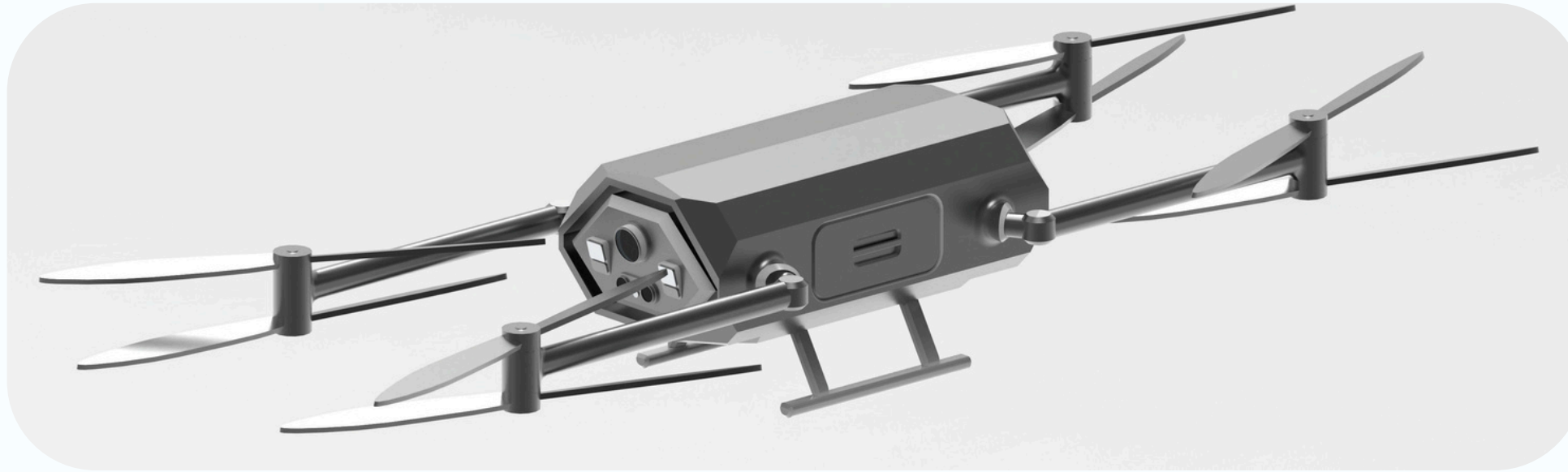
- Actuation: 8-rotor coaxial octocopter
- Connectivity: Long-Range LoRa technology
- Sensors: Dual 360° LiDAR + UWB Radar

Core AI capabilities:

- Sensor Fusion: Spot hidden survivors in harsh environments by fusing LiDAR, UWB Radar, and Thermal Camera data.
- Load Balancing: Prevent pendulum effect during extraction by automatically adjusting motor thrust.

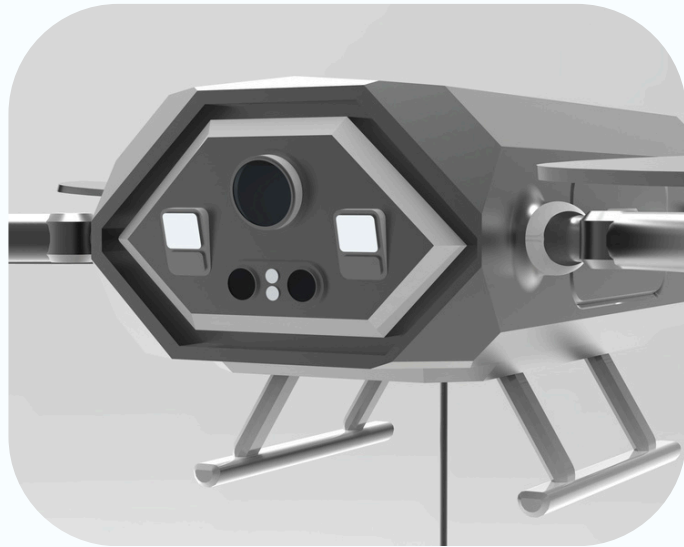


A.R.E.D. | Structural Architecture Design



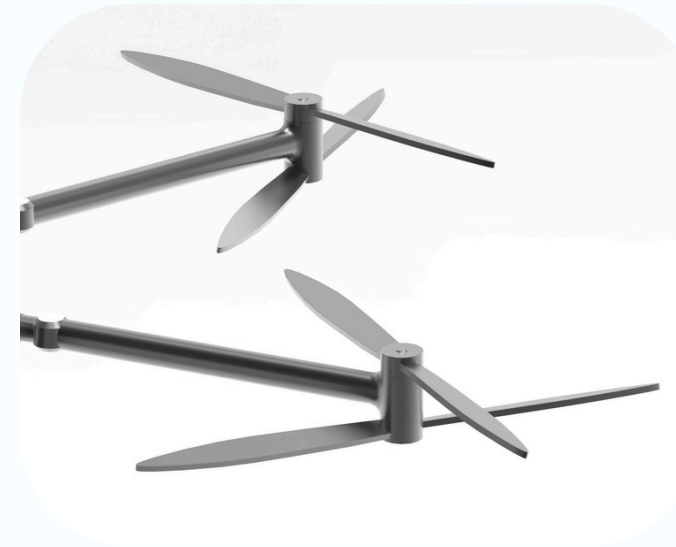
Foldable Arm Hinge System

- Heavy-duty articulated mechanical joints allow the arms to be folded for easy storage and transport.



Streamlined Chassis

- Carbon fiber enclosure with chamfered edge to reduce wind resistance.



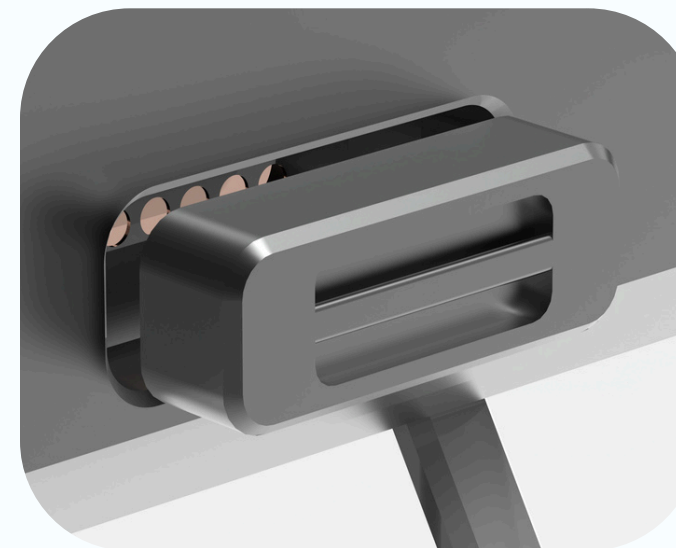
8-Rotor Coaxial System

- Coaxial dual-motor layout provides heavy lifting for human harness.



Dual Skid Landing Gear

- Wide landing gear skid railing to securely land on uneven surface without tipping over.



Battery & Electrical Bay

- Side-mounted modular bays for instant battery and flight controller replacement.

A.R.E.D. | Victim Extraction Design

1

Safe Winch System & D-Ring Coupling

- The drone centre winch lowers an adjustable 15 - 20 meter high-strength cable coupled with a heavy-duty D-ring connector to secure the victim through tree canopies or debris.

2

Full-Body Support Harness

- The ergonomic full-body safety harness prevents injury during lifting and extraction.

3

6.3-Inch Diagnosis Screen

- The built-in diagnosis screen provides the victim immediate paramedic attention using long-range LoRa technology to establish a 2-way video/audio connection with ground doctors.

4

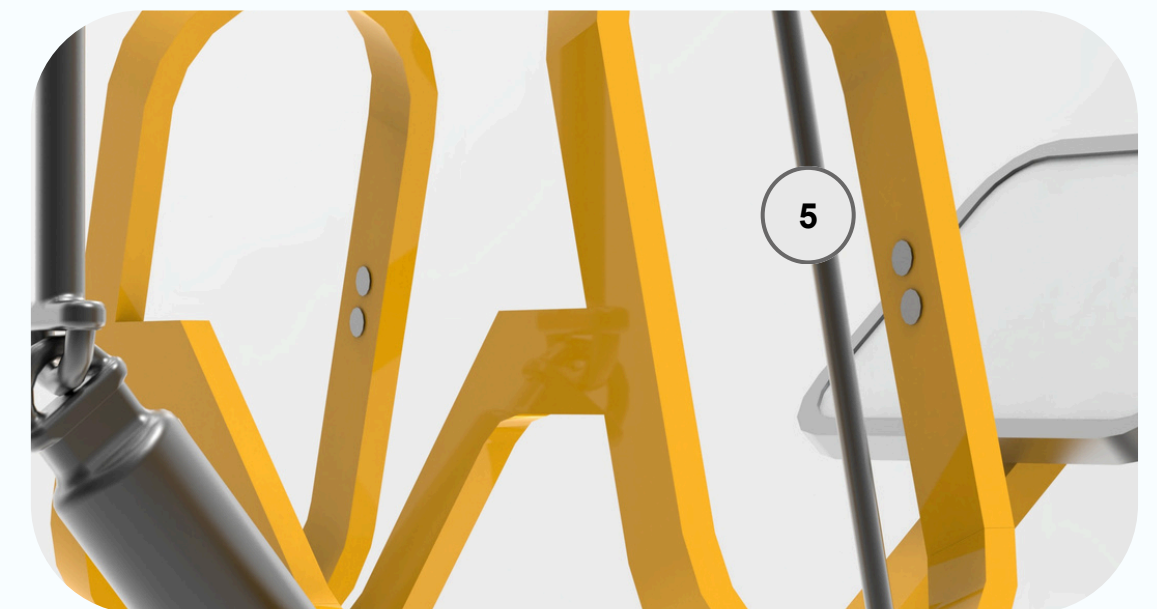
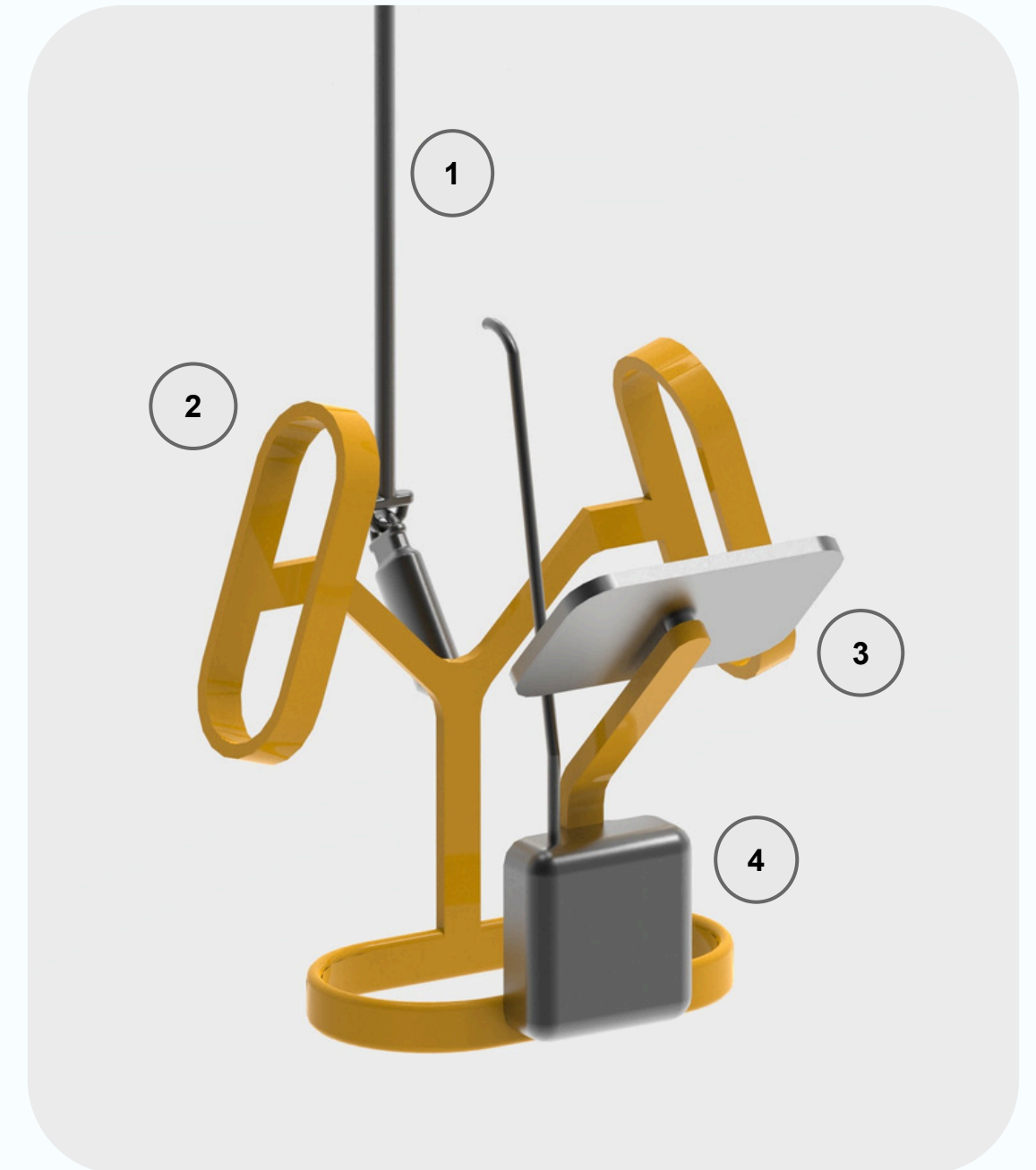
Built-in Hydration Pack

- The built-in water pack replenishes the dehydrated victim's hydration immediately with a hand-free straw.

5

Biometric Mesh Sensor

- The integrated biometric sensors monitor real-time heart rate and oxygen saturation (SpO₂) for immediate diagnosis.



A.R.E.D. | AI Capability Concept

Autonomous Flight & Load Balancing

- The onboard AI continuously calculates real-time wind speed and load weight to automatically adjust each motor thrust to cancel out the rope swing pendulum effect and ensure extraction safety.

AI Sensor Fusion

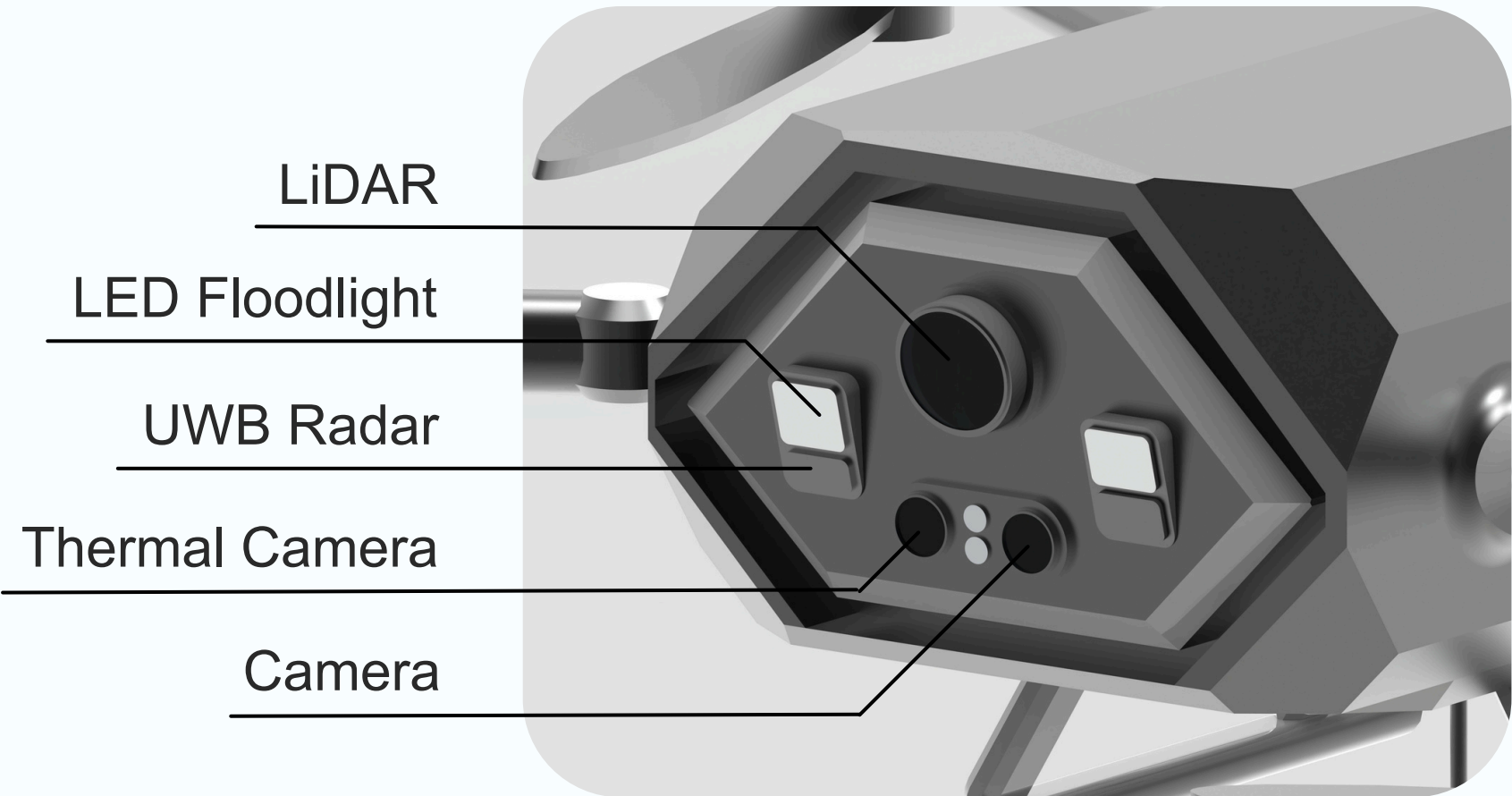
- The AI fuses real-time data from the dual 360° LiDAR, UWB Radar, and Thermal Imaging to spot human heartbeats and body heat through thick smoke, water, or collapsed rubble.

Biometric Health Monitoring

- The AI analyses the victim’s heart rate and oxygen saturation (SpO₂) from the harness mesh to automatically detect health conditions and transmit alert through LoRa to the first responders for urgent medical attention.

Interactive AI Voice Assistant

- The smart harness screen uses AI voice to communicate with victims to comfort them, provide alerts, and guide them on how to secure the harness.



Signal Flow Diagram

